Εισαγωγή στη Φωτογραμμετρία
Ομάδα ασκήσεων 4

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Άσκηση 1

$$\begin{array}{c} \begin{array}{cc} K_{11} & K_{12} \end{array} \\ \left[\begin{array}{cc|cc} 10 & 2 & 3 & 5 \\ 6 & 5 & 1 & 2 \\ \hline 4 & 9 & 11 & 3 \\ 15 & 20 & 8 & 7 \end{array}\right] \\ \begin{array}{cc} K_{21} & K_{22} \end{array} \end{array} \qquad \begin{array}{c} l_1 \\ \left[\begin{array}{cc} 20 & 17 \\ 23 & 21 \\ \hline 16 & 15 \\ 19 & 18 \end{array}\right] \\ l_2 \end{array}$$

Τι μας έδωσε ο καθηγητής

$$\begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} = \begin{bmatrix} K_{11} \cdot l_1 + K_{12}\, l_2 \\ K_{21} \cdot l_1 + K_{22}\, l_2 \end{bmatrix}$$

α) Το δικό μας:

$$\begin{array}{c} \\ K_{11} \\ K_{21} \end{array} \begin{array}{c} \begin{array}{cc} & K_{12} \end{array} \\ \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 1 & 4 \\ \hline 2 & 2 & 3 \end{array}\right] \\ \begin{array}{cc} & K_{22} \end{array} \end{array} \begin{array}{c} \begin{array}{cc} l_{11} & l_{12} \end{array} \\ \left[\begin{array}{cc|c} 0 & 3 & 1 \\ 4 & 2 & 1 \\ \hline 3 & 5 & 2 \end{array}\right] \\ \begin{array}{cc} l_{21} & l_{22} \end{array} \end{array} = \begin{bmatrix} 2 & 12 & 3 \\ 20 & 38 & 18 \\ 17 & 39 & 10 \end{bmatrix}$$

$$\begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \cdot \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{bmatrix} = \begin{bmatrix} K_{11} \cdot l_{11} + \overset{①}{K_{12}} \cdot l_{21} & K_{11} \cdot l_{12} + \overset{②}{K_{12}} \cdot l_{22} \\ K_{21} \cdot l_{11} + \underset{③}{K_{22}} \cdot l_{21} & K_{21} \cdot l_{12} + \underset{④}{K_{22}} \cdot l_{22} \end{bmatrix}$$

$$① \rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 0 \cdot 4 & 1 \cdot 3 + 2 \cdot 2 \\ 3 \cdot 0 + 2 \cdot 4 & 3 \cdot 3 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 8 & 18 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 0 \cdot 3 & 0 \cdot 5 \\ 4 \cdot 3 & 4 \cdot 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 12 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 12 \\ 8 & 18 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 12 & 20 \end{bmatrix} = \begin{bmatrix} 2+0 & 12+0 \\ 8+12 & 18+20 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 20 & 38 \end{bmatrix} \Big\} \; 1$$

$$2 \rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 2\cdot 1 \\ 3\cdot 1 + 1\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 4 \end{bmatrix} [2] = \begin{bmatrix} 0 \\ 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \end{bmatrix} \Big\} \; 2$$

$$3 \rightarrow [2 \;\; 2] \begin{bmatrix} 0 & 3 \\ 4 & 2 \end{bmatrix} = [2\cdot 0 + 2\cdot 4 \;\; 2\cdot 3 + 2\cdot 2] = [8 \;\; 24]$$

$$[3]\cdot[3\cdot 5] = [9 \;\; 15] \Rightarrow [8 \;\; 24] + [9 \;\; 15] = [17 \;\; 39] \Big\} \; 3$$

$$4 \rightarrow [2 \;\; 2] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = [2\cdot 1 + 2\cdot 1] = [4]$$

$$[3]\cdot[2] = [6] \Rightarrow [4] + [6] = [10] \Big\} \; 4$$

Τελικό αποτέλεσμα →

$$\begin{array}{cc} 1 & 2 \\ \end{array}$$
$$\left[\begin{array}{cc|c} 2 & 12 & 3 \\ 20 & 38 & 12 \\ \hline 17 & 39 & 10 \end{array}\right]$$
$$\begin{array}{cc} 3 & 4 \end{array}$$

β) Άλλος τρόπος:

$$\begin{array}{cc} K_{11} & K_{12} \\ \end{array} \qquad \begin{array}{cc} \lambda_{11} & \lambda_{12} \\ \end{array}$$
$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 3 \\ 5 & 3 & 2 & 4 \\ \hline 4 & 1 & 0 & 2 \end{array}\right] \left[\begin{array}{c|c} 5 & 0 \\ 3 & 2 \\ \hline 3 & 0 \\ 3 & 1 \end{array}\right] = \begin{bmatrix} 23 & 7 \\ 52 & 14 \\ 29 & 4 \end{bmatrix}$$
$$\begin{array}{cc} K_{21} & K_{22} \\ \end{array} \qquad \begin{array}{cc} \lambda_{21} & \lambda_{22} \\ \end{array}$$

$$A_{3\times 4} \times B_{4\times 2} = C_{3\times 2}$$

$$1 \rightarrow \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1\cdot 5 + 2\cdot 3 \\ 5\cdot 5 + 3\cdot 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 34 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1\cdot 3 + 3\cdot 3 \\ 2\cdot 3 + 4\cdot 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 18 \end{bmatrix} \Rightarrow \begin{bmatrix} 11 \\ 34 \end{bmatrix} + \begin{bmatrix} 12 \\ 18 \end{bmatrix} = \begin{bmatrix} 23 \\ 52 \end{bmatrix} \Big\} \text{①}$$

$$\text{②} \rightarrow \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1\cdot 0 + 2\cdot 2 \\ 5\cdot 0 + 5\cdot 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 0 + 3\cdot 1 \\ 2\cdot 0 + 4\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 14 \end{bmatrix} \Big\} \text{②}$$

$$\text{③} \rightarrow \begin{bmatrix} 4 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 4\cdot 5 + 1\cdot 3 \end{bmatrix} = \begin{bmatrix} 23 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 0\cdot 3 + 2\cdot 3 \end{bmatrix} = \begin{bmatrix} 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 23 \end{bmatrix} + \begin{bmatrix} 6 \end{bmatrix} = \begin{bmatrix} 29 \end{bmatrix} \Big\} \text{③}$$

$$\text{④} \rightarrow \begin{bmatrix} 4 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 4\cdot 0 + 1\cdot 2 \end{bmatrix} = \begin{bmatrix} 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0\cdot 0 + 2\cdot 1 \end{bmatrix} = \begin{bmatrix} 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix} = \begin{bmatrix} 4 \end{bmatrix} \Big\} \text{④}$$

Τελικό
Αποτέλεσμα →

$$\begin{array}{cc} \text{①} & \text{②} \\ \left[\begin{array}{c|c} 23 & 7 \\ 52 & 14 \\ \hline 29 & 4 \end{array}\right] \\ \text{③} & \text{④} \end{array}$$